

MELISA OSHAINI ARUKGODA

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Objective

Motivated researcher with a strong interest in Robotics and Machine Learning, eager to advance knowledge and develop innovative approaches for intelligent robotic systems through rigorous, high-impact research.

Experience

Research Assistant - Bio-Inspired Robotics

Feb 2026 - Present

Pennsylvania State University, Dept. of Mechanical Engineering (Onsite)

- Research under Prof. Baxi Chong on bio-inspired robot locomotion.
- Focusing on modeling, simulation, and optimization of geometric locomotion.

Geometry Processing Researcher - Summer Geometry Initiative

July 2025 – Aug 2025

Massachusetts Institute of Technology (Remote)

- **Motion planning for broken snake robots** – Explored geometric strategies for fault-tolerant motion planning in shape-changing robots.
- **Penetration-free deformation field computation** – Implemented algorithms to compute deformation fields while avoiding self-penetration in 3D shapes.
- **Topological analysis of AI-generated images** – Analyzed structural patterns and stability of AI-generated visuals using topological methods.

Intern Research Assistant

Dec 2023 – June 2024

Robotics and Automation Research Laboratory, SUTD, Singapore (Onsite)

- Worked on the project “Optimum Placement of LiDAR Sensors for Autonomous Robots” under Prof. Mohan Rajesh Elara.
- Developed an educational simulation platform using Unity and a mobile app to control a reconfigurable robot.

Visiting Instructor

2023

University of Moratuwa

- Conducted practical lab sessions for EN2553: Robot Design and Competition.

Education

University of Moratuwa, Sri Lanka

Mar 2021 – June 2025

B.Sc. Honors in Engineering, Minor in Mathematics

CGPA: 3.82/4.00

Department of Electronic and Telecommunication Engineering

Dean's List: Semester 1,2,4,6,7,8

Ferguson High School, Ratnapura, Sri Lanka

Jan 2011 – Dec 2019

Physical Science Stream

Z score: 3.7177

GCE Advanced Level(Mathematics,Physics,Chemistry,General English) *4As/ 74th in the Country*

(country-wide university entrance examination taken by over 100,000 students annually)

Other Courses

Courses followed to self-learn the content required for the projects outside the full-time degree.

- Machine Learning: On Coursera
- Computer Vision: On Kaggle
- ROS for Beginners: Basics, Motion & OpenCV: On Udemy

Publications

Manuscripts Under Review

- **Melisa Oshaini Arukgoda**, Bhagya Samarakoon, M. A. Viraj J. Muthugala, Mohan Rajesh Elara. **Determining the Optimum Placement of LiDAR Sensors for Autonomous Robots.**
Under review at IEEE Access Journal
- **Melisa Oshaini Arukgoda**, Charuka Bandara, Tharusha Fonseka, Moshintha Isuru Hewavitharana, Wageesha N. Manamperi, Udaya S. K. Perera Miriya Thantrige, Peshala Jayasekara. **Articulated Humanoid Head for a Robot Receptionist Capable of Natural Human Interaction.**
Under review at HRI 2026

Projects

- Final Year Project – Kuweni: Humanoid Robot Receptionist** | [Project URL](#) **Aug 2024 – June 2025**
- Developed a 21-DOF humanoid robotic head capable of expressing six basic emotions via silicone skin and precision actuation, aimed at natural and engaging human-robot interactions.
 - Designed and 3D printed mechanical subsystems including eyes, eyebrows, jaw, and neck. Implemented control logic using the ROS2 framework.
 - Integrated AI modules for multi-user facial recognition (89.7% accuracy), speech recognition (11.43% WER), and sound localization (1.79° error using GCC-PHAT).
 - Achieved a 4.91s average response time using LLaMA 3.1 and Retrieval-Augmented Generation (RAG). Evaluated with 60 users for interaction quality (4.3/5.0) and expression realism (94.2%).
- Speech Emotion Recognition (Wav2Vec2 + LSTM)** | *PyTorch, Hugging Face, Model Optimization* [URL](#) **Aug 2025**
- Achieved 97% validation accuracy on TESS using PyTorch with Wav2Vec2 embeddings, LSTM/bi-LSTM, and data augmentation.
- Maze Solving Micromouse Robot (Robofest)** | *Python, STM32CubeIDE* [Git](#) **Sep 2024**
- Built a fully Autonomous maze solving robot, which is tasked to reach the destination point in complex mazes in the shortest time possible.
 - The robot had floodfill algorithm-based navigation and STM32 microcontroller-based hardware implementation.
- Ophthalmic Biomarker Detection (IEEE VIP Cup 2023 - Ranked 6th Globally)** | *Pytorch* [Git](#) **Aug 2023**
- Optimized the algorithm to predict the presence or absence of 6 different biomarkers simultaneously on OCT scan images and tested with different preprocessing techniques and backbones and optimized the model by changing the classification layers.
- Dual Mode Robot Mop** | *Altium Designer, Solidworks, Arduino, Mobile App, Product Design* [Git](#) **Apr – Jul 2023**
- Individual project: Designed and developed a rechargeable robot mop with both autonomous and manual modes, incorporating robotics, product design, and problem-solving.
- Battle Bot - Scorpion (EXMO 2023)** | *Solidworks, C++* [Git](#) **Jul 2023**
- Designed a battle robot to demonstrate in EXMO 2023. This includes a motorized hook and a linear roller, which were controlled by Arduino Mega.
- Mobile App Development for Reconfigurable Shape-Changing Robot Controller** | *Flutter* **Dec 2023 - Jan 2024**
- Developed a Flutter-based mobile [app](#) with low-energy Bluetooth for controlling a reconfigurable shape-changing robot, deployed on both iOS and Google Play Store.
- Analog Function Generator** | *Altium Designer, Electronic Circuit Design* [Git](#) **Oct 2022 – Mar 2023**
- Developed an Analog Function generator that can generate a sine wave, square wave, sawtooth, and triangular wave with variable amplitude. The circuit was designed using op-amps to become fully analog. Op-Amps were optimally manipulated by not just limiting to conventional wave-generating models. So we just needed six op-amps and were able to design the PCB so tiny with the best PCB designing practices.
- EcoQuest: Unity Based Energy Conservation Game Development** | *Unity, Firebase* [Git](#) **Mar – Jun 2024**
- Developed a Unity-based 2D game that educates user on energy conservation through interactive gameplay, highlighting the impact of energy saving decisions.
- Smart Extension Socket** | *Solidworks, Electronic Circuit Design* [Git](#) **Jun – Sep 2022**
- Led the successful development of a 'Smart Extension Socket' project, effectively addressing energy conservation challenges with user-friendly timer functionality.

Technical Skills

Languages: Python, MATLAB, C#, Dart

Software & Tools: ROS2, Pytorch, Hugging Face, Altium Designer, Solidworks, Webots, Unity, Flutter(App Development)

Others: L^AT_EX, Arduino, Docker

Relevant Coursework

Robotics: EN2533 Robot Design and Competition , EN3563 Robotics, EN2143 Electronic Control Systems, EN4594 Autonomous Systems, EN4660 Advanced Electronic Control Systems

Computer Vision: EN3160 Image Processing and Machine Vision, EN4554 Deep Learning for Vision

Mathematics: MA2024 Calculus , MA2034 Linear Algebra , MA2014 Differential Equations , MA1014 Mathematics(Real Analysis) , MA1024 Methods of Mathematics(Numerical Methods) , MA3014 Applied Statistics, MA4014 Linear Models and Multivariate Statistics, MA4144 Neural Networks and Fuzzy Logic

Achievements and Awards

MIT SGI 2025: Selected as one of the 30 fellows worldwide for MIT's Summer Geometry Initiative	2025
Finalist: Robofest Micromouse Competition organized by Sri Lanka Institute of Information Technology	2024
Ranked 600th globally out of 8000+ teams: IEEE Xtreme	2024
Finalist: Robofest Micromouse Competition organized by Sri Lanka Institute of Information Technology	2023
Ranked 6th globally: IEEE Video and Image Processing Cup	2023
Finalist: SPARK Challenge organized by SPARK Foundation and University of Moratuwa	2023
Mahapola Merit Scholarship: Ranked 74th in Sri Lanka at GCE Advanced level Examination	2019
Gold Medalist: All Island Mathematics Competition, Sri Lanka	2018
Second Best Performance in Senior 2 Division: Australian National Chemistry Quiz	2018
High Distinction: Sri Lanka Mathematical Olympiad	2018
High Distinction: Sri Lanka Mathematical Olympiad	2017

Volunteer Experience/ Leadership

Secretary: Electronic Club(SPARK Branch) of University of Moratuwa	Aug 2024 – July 2025
Judging Panel Member: RoboRoarz Singapore 2024	Mar 2024
Organizing Committee Member: RoboRoarz Singapore 2024	Mar 2024
Project Manager: Electronic Club(SPARK Branch) of University of Moratuwa	Aug 2023 – Aug 2024
Assistant Academic Project Manager: Electronic Club of University of Moratuwa	Jul 2022 – Aug 2023
President: Mathematics and Science Society of Ferguson High School	Dec 2016 – Dec 2017
Junior Prefect: Ferguson High School	Dec 2014 – Dec 2015
Member: Drama and Theatre Club of Ferguson High School	Dec 2014 – Dec 2015

Soft Skills

Communication Skills: Languages

- English: Professional Working Proficiency
- Sinhalese: Native Proficiency

Leadership and Teamwork:

- Experienced in leading and working with multiple teams and clients as the Secretary of Electronic Club (SPARK branch).
- Project group leader of undergraduate projects.
- Experienced in collaborating with international teams on projects.

References

DR VIRAJ MUTHUGALA

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DR PESHALA JAYASEKARA

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